

**I Wish I Was Blue!: The Development of (Un)attainable Desires
in Child Greek**

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1. Introduction

Across languages, we use propositional attitude reports to describe the object of desire, i.e. clause embedding sentences built around attitude verbs like *want*, *wish* and *hope* (Russel 1940, Grano 2021, Pearson 2020). When we use an expression like (1) we express a desire about something we want to do or to be; whether this desire is possible to be fulfilled or not in this world is somehow conveyed by grammatical cues. Thus, the complement of (1a) (*to plant a carob tree*) is a future oriented desire, possible to be attained in this world, but the complement of (1b) (*(s)he had planted a carob tree*) is a wish, i.e. a counterfactual desire which is not attainable in the real world.

- (1) a. Theli na fitepsi mia harupça. [O-marked desire]
 want.NPST.IPFV na.SUBJ plant.NPST.PERF a carob-tree
 ‘(S)he wants to plant a carob tree.’
 b. Tha ithele na ihe fitepsi mia harupça. [X-marked desire]
 want.PST.IPFV na.SUBJ plant.PLUP a carob-tree
 ‘(S)he wishes (s)he had planted a carob tree.’

Up-to-date, the development of desires and wishes has received limited attention experimentally across languages. In this study we aim at shedding light on the development of desires and wishes by testing children’s comprehension of attainable and unattainable (counterfactual) desires and by comparing children’s comprehension on counterfactual desires to counterfactual conditionals. Following von Fintel & Iatridou’s (2023) terminology, we will refer to future oriented, attainable desires which are open to fulfilment (1a) as *O-marked desires*

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and to counterfactual, unattainable desires which cannot be fulfilled in the actual world (1b) as *X-marked desires*.¹

We show that children’s understanding of O-marked desires and conditionals is adultlike while their interpretation of X-marked desires and conditionals reveals a prolonged developmental process of counterfactuality: children up to the age of 7 interpret X-marked desires and conditionals as O-marked at a large rate. We believe this difficulty stems from a misanalysis of the pluperfect form which is associated with X-marking and also from children’s Actuality bias, i.e. the bias towards reality in verb learning. (de Villiers 2005, Kazanina, Baker & Seddon 2020).

2. Desires and wishes in Modern Greek

The form languages employ to express wishes typologically distinguishes them into transparent and non-transparent wish-languages (von Fintel & Iatridou 2023). According to the authors, transparent wish-languages (e.g. Greek, Spanish, Farsi, Hungarian) are typically the ones in which a wish is expressed via morphological X-marking on the desire predicate *want*. Non-transparent wish-languages are characterized by having a *wish* dedicated word (e.g. English, Turkish, Burmese, Hindi, Croatian).

In our view, Modern Greek belongs to both transparent and non-transparent languages as it employs both the X-marked *want*-form and verbs like *wish* (*efhome*) to express wishes². Moreover, the difference between O-marked and X-marked desires in MG is conveyed through obligatory morphological marking on the verb of the complement clause (contra von Fintel & Iatridou 2023), whereas the desire-predicate itself does not necessarily have to be morphologically X-marked. The morphological marking of the complement verb in (2b-c) is compatible with both the forms *theli* and *tha ithele*³; the X-marked complement

¹ Throughout our paper we use the term X-marked desire to refer to unattainable desire, agreeing with von Fintel and Iatridou (2023) that the term ‘counterfactual wish’, which is often used in the literature, is potentially misleading as the desire expressed in (1b) still holds in the actual world and not in a counterfactual world; only the complement of the desire/wish is not possible to be realised. This becomes particularly relevant in languages like Greek which has the form of what we would call a ‘counterfactual wish’ as in (i):

(i) Tha efxomun na ixa spudasi fisiki.

MOD wish-1SG.PST.IPFV na.SUBJ study-1SG.PLUP physics

‘I wish I had studied physics.’

The meaning in (i) is equivalent to ‘if I had a wish (but I do not have one), I would want to have studied physics’.

² Note that the authors themselves point out in a footnote that this classification is actually a misleading simplification if one looks into the lexical properties of the verbs within a language. See also Longenbaugh’s (2019) proposal regarding the *non-logical-if-constructions* being simply X-marked desire reports.

³ Note that while (2b-c) are grammatical with present form on the desire verb *want* (i.e. *theli*), they are more context dependent, as for example in answering a question like “why

Thus, when the complement of the verb is in present tense (perfective or imperfective), the sentence is O-marked (2a). When the complement of the verb is in pluperfect (2b), the utterance is pastX-marked and when the complement of the verb is in past imperfective (2c), the utterance is either presentX-marked or O-marked, if it refers to a scenario that is less possible to be fulfilled in the future (Future Less Vivid, Iatridou 2000). The morphological marking on the desire-verb itself is contingent on the verb-type, as shown by the past imperfective of *thelo* ('want') in (2b-c) in comparison to the unvarying *efhomo* ('wish') in (2a-c).

(2) a. Theli/efçete na fitepsi mia harupça.
want_{3SG.NPST.IPFV} / wish_{3SG.NPST.IPFV} na_{SUBJ} plant_{NPST.PERF} a carob-tree
'(S)he wants to plant a carob tree.'

b. Tha ithele/ ?theli / efçete na içe fitepsi mia harupça.
MOD want_{3SG.PST.IPFV/NPST.IPFV}/wish_{3SG.NPST.IPFV} na_{SUBJ} plant_{PLUP} a carob-tree
'(S)he wishes (s)he had planted a carob tree.'

c. Tha ithele/ ?theli /efçete na fiteve mia harupça.
MOD want_{3SG.PST.IPFV/NPST.IPFV}/wish_{3SG.NPST.IPFV} na_{SUBJ} plant_{PST.IPFV} a carob-tree
'(S)he wishes (s)he would plant a carob tree.'

(3) a. Makari/ax ce/tulaxiston na spudazi fisiki
 MARKER na.SUBJ study.3SG.NPST.IPFV physics
 ‘If only he studies physics.’
 b. Makari/ax ce/tulaxiston na ixe spudasi fisiki

⁴ We wish to report that strictly speaking there is a fourth class which includes verbs expressing ‘dislikeness’, e.g. *miso* (hate), *siçenome* (despise), and *apeşhime* (wish against), which do not uniformly fit in one class, and they are very infrequently used in X-marked sentences. However, we will not say more on this as it goes beyond the scope our paper.

MARKER na.SUBJ study.3SG.PLUP physics
 ‘If only he had studied physics.’

Unlike embedded wishes, examples in (3) are performative utterances in which the attitude holder is invariably the speaker and the wish can only be evaluated at context time (see e.g. Condoravdi & Lauer, 2012, Oikonomou 2016). For this reason we did not include them for testing potential differences between O-marked and X-marked desires.

3. The similarity of wishes to X-marked conditionals in MG

Across languages, there is a morphological similarity between X-marked desires and X-marked conditionals, formulated as the Conditional/ Wish pattern (4) (von Fintel & Iatridou 2023). The complement of the desire-verbs in X-marked desires and the antecedent of counterfactual conditionals appear with the same morphology (Iatridou 2000).

(4) The Conditional/ Wish pattern (von Fintel & Iatridou 2023: 1485)

X-marked conditional: if p_{ANTECEDENT} then q_{CONSEQUENT}

X-marked desire: I want_{CONSEQUENT} that p_{ANTECEDENT}

Thus, the X-marked desire-verb (*tha ithela* in (1b)), has the same morphology as the verb of the consequent of the conditional in (5). Similarly, the complement of the desire-verb in X-marked desires and the antecedent of an X-marked conditional, appear with the same morphology (cf. (1b) to (5a)). Pluperfect or past imperfective is used in the complement of the desire predicate in (2b) and (2c) respectively and in the antecedent of the conditional in (5a) and (5b) respectively.

(5) a. An içe fitepsi mia harupça tha itan harumeni.

If plant_{PLUP} a carob-tree MOD be_{.PST,IPFV} happy_{FEM}

‘If she had planted a carob-tree, she would be happy.’

b. An fiteve mia harupça tha itan harumeni.

If plant_{PST,IPFV} a carob-tree MOD be_{.PST,IPFV} happy_{FEM}

‘If she would plant a carob-tree, she would be happy.’

The similarities between the two constructions (2b) and (5a), both expressing counterfactuality, is the reason we compare their development in Experiment II.

4. Previous studies on the development of desires and wishes

It has been suggested that in the development of the attitude verbs, children acquire desire-verbs before belief-verbs (Perner, Sprung, Zauner & Haider 2003) and the development of the “desire reasoning” proceeds the development of “belief reasoning”; at about the age of two years, children correctly predict actions related to simple desires while they fail with belief tasks (Wellman & Wooley

1990). Moreover, Repacholi & Gopnik (1997) have claimed that already at the age of 18-months, children can understand desires that are different from their own. However, Moore, Jarrold, Russell, Lumb, Sapp & MacCallum (1995) have claimed that three-year-olds can judge the desire of another person correctly only if they, themselves, do not hold a strong conflicting desire, and that only five-year-olds can understand that a person can desire something different from them. The latter was challenged by Harrigan, Hacquard & Lidz (2018) who suggested that three-year-olds can understand desires that conflict with their own desires and with reality, showing that children have an adultlike understanding of the verb *want* already by the age of three. It should be noted that in the study of Harrigan, Hacquard & Lidz (2018), their counterfactual desires are tested via scenarios which conflict with reality and they are not expressed as linguistic counterfactual structures. Moreover, in their examples it is uncertain whether there is a clear conflict with reality or a mismatch between conflicting desires. It is then unclear whether success on these tasks reflects a fully-fledged concept of counterfactual desires and their linguistic characteristics.

Evidence regarding the development of X-marked desires comes from the studies of Tulling & Cournane (2019, 2022), which were based on naturalistic longitudinal data (CHILDES database) from English speaking children from 2-6 years old. They found that children often produce counterfactual constructions that lack the ‘fake’ past tense, i.e. the pluperfect on the complement verb and they make productive present-for-past errors at the age of 2-4 years. The authors also suggested that children produce X-marked desires before X-marked conditionals in English.

Understanding counterfactual conditionals and counterfactual desires is a demanding task for children as it requires the development of counterfactual thinking in combination with a complex form-to-meaning mapping (Rouvoli, Tsakali & Kazanina 2019, Tulling & Cournane 2022). In the Rouvoli, Tsakali & Kazanina (2019) study, the findings suggest that children’s failure with past counterfactuals stems from their lack of understanding the linguistic form, namely that *if* + *pluperfect* denotes counterfactuality, and not from their inability to think counterfactually.

Beck, Robinson, Carroll & Apperly (2006) explored children’s understanding of future hypothetical conditionals and counterfactuals. They suggested that children understand future hypotheticals correctly independently from their ability to entertain alternative worlds, but they are cognitively mature to master counterfactuals only after preschool years, when they have developed the ability to think on possible worlds. Along this line, it has been suggested that before the age of six, children use basic conditional and not counterfactual reasoning (Rafetseder, Christi-Vargas & Perner 2010) and that counterfactual reasoning does not develop fully before the age of 12 years (Rafetseder, Schwitalla & Perner 2013). Younger children’s inability to interpret counterfactuality is also attempted to be explained by the Actuality bias hypothesis, according to which children first learn the syntactic and semantic constructions that are related to reality (de Villiers 2005; Kazanina, Baker & Seddon 2020).

In general, there is agreement in the literature that the interpretations of sentences that are not in conflict with reality are acquired first, explaining why children have difficulty to differentiate between real and unreal information (Gómez-Sánchez, Ruiz-Ballesteros & Moreno-Ríos 2020).

To the best of our knowledge, the development of counterfactual desires has not been studied experimentally and the question whether counterfactual conditionals and counterfactual desires develop simultaneously remain open. Testing the development of the linguistic aspects of future oriented (O-marked) desires and conditionals in comparison to counterfactual (X-marked) desires and conditionals is necessary to explore how desires and counterfactuality develop.

5. The learnability questions

The study aims at exploring when and how children acquire O- and X-marked desires. It further aims to investigate the development of X-marked desires in comparison to X-marked conditionals. The main questions this study attempts to shed light on are:

- (i) At what age do children start distinguishing between O-marked and X-marked desires in MG?
- (ii) What are the stages children go through and what types of errors describe each stage, prior to reaching adult-like performance?
- (iii) Does the development of counterfactuality across structures vary? In other words, does the development of X-marked desires proceed or follow the development of X-marked conditionals?

The first question, besides giving evidence regarding the developmental path, will ultimately provide crucial evidence regarding potential differences on the crosslinguistic picture. The second question aims at drawing our attention to the types of errors which are characteristic to each stage, to the extent that this is detectable, and finally, the third question is expected to enrich our understanding as to whether the acquisition of X-marked desires deviates from that of X-marked conditionals.

6. Experiment I: Comparing O-marked to X-marked desires

6.1. Task and procedure

A comprehension picture-matching task in PCIBeX (Zehr & Schwarz 2018) was developed to test children's understanding of O-marked and X-marked desires.

The targeted question was framed in a 'Tell-me-who-is-talking' mode: children were told a short story about little creatures who went to the Festival of Desires, where they can become any colour they want. Children are informed that they will listen to the 'interviews' the little creatures gave afterwards, talking about the colour they got and saying whether they want to remain the same or if they would like to be different. Children would watch two little creatures on a screen (as in Fig.1 and Fig.2), while listening to a recorded utterance (as in 6a-d).

They were instructed from the beginning that they would have to tell the experimenter which creature is talking after listening to the utterance and click on the icon in order to proceed to the next picture. The experimenter's question was: *pjo plasmataki milise (tora)?* ('Which little creature talked (now)?').

Children were also informed that they can listen to the utterance again if they wanted to and we made sure all children know the colours mentioned in the experimental procedure (i.e. red, blue, purple, yellow, pink, orange, brown and green).

There were 3 warm up items for children's familiarization with the procedure, followed by 24 critical items (6 items/condition) and 24 fillers. All items appeared in random order per participant; the order of the 'speaking' creature in the picture also appeared counter-randomly on the right or the left side of the screen.

The duration of the study was 12-17 minutes and after completion, the experimenter thanked children for participating and rewarded them with stickers.

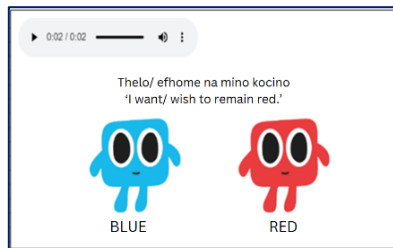


Figure 1: Example of O-marked desire (Condition 1 & 2)

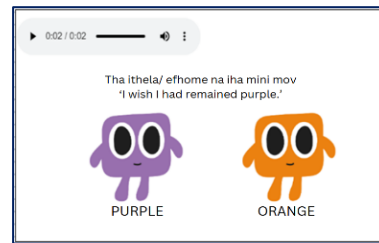


Figure 1: Example of O-marked desire (Condition 3 & 4)

6.2. Conditions and Items

Experiment I contains four conditions (6 items/ condition) and aims at testing potential differences between: a) the two desire-verbs, namely *thelo* (want) and *efhome* (wish) and, b) O-marked and pastX-marked desires. The choice for these two desire-verbs aimed at examining whether children are sensitive to the presence of X-marking morphology on the desire-verb itself. Recall that *efhome* (wish) is obligatorily unchangeable in O-marked and X-marked structures (see discussion in Section 2).

- (6) **Condition 1: O-marked *thelo* ('want')** [6 items]
 a. Thelo na mino kocino.
 want._{1SG.NPST.IPFV} na._{SUBJ} remain._{1SG.NPST.PERF} red
 'I want to remain red.'
- Condition 2: O-marked *efhome* ('wish')** [6 items]
 b. Efhome na mino kocino.
 wish._{1SG.NPST.IPFV} na._{SUBJ} remain._{1SG.NPST.PERF} red
 'I wish to remain red.'
- Condition 3: X-marked *thelo* ('want')** [6 items]

c. Tha ithela na iha mini mov.
 MOD want.1SG.NPST.IPFV na.SUBJ remain.PLUP purple
 ‘I wish I had remained purple.’

Condition 4: X-marked *efxome* (‘wish’) [6 items]

d. Efxome na iha mini mov.
 wish.1SG.NPST.IPFV na.SUBJ remain.PLUP purple
 ‘I wish I had remained purple.’

Additionally, we used 24 control-fillers which were designed uniformly to the critical items of the experimental task. The fillers intended to shed light on how children respond to the desire-verb *aresi* (like) as it has been observed that it appears early in child language and it is more frequent in the input. Moreover, the verb *aresi* belongs to Class III (See Section 2), which is obligatorily morphologically marked with past imperfective in X-marked conditions and we wanted to explore the possibility whether such a linguistic cue may affect children’s performance. Finally, the fillers’ design intended to control for potential strategies in O-marked conditions. Thus, the verb *aresi* appeared in two O-marked contexts (Condition 5 & Condition 7; examples (7a) and (7c) as depicted in Fig.3) and in two X-marked contexts (Condition 6 & Condition 8; examples (7b) and (7d) as depicted in Fig.4). The verb of the complement clause included activity verbs (e.g. *petao* (fly)) (7a-b) and *be*-predicate (7c-d) in O-marked and X-marked morphology (past imperfective), which is FLV without a context, but it receives counterfactual meaning in the specific experimental context.



Figure 3: Example of fillers with activity verb complement/ O-marked & X-marked (Condition 5 & 6)



Figure 4: Example of fillers with be-predicate verb complement/ O-marked & X-marked (Condition 7 & 8)

(7) Condition 5: Like/ O-marked activity [6 items]

a. Mu aresi pu petao.
 me.DAT like.NPST that fly.1SG.NPST.IPFV
 ‘I like that I fly.’

Condition 6: Would like/ X-marked activity [6 items]

b. Tha mu arese na petusa.
 MOD me.DAT like.PST.IPFV that fly.1SG.PST.IPFV
 ‘I would like to fly.’

Condition 7: Like/ O-marked predicate [6 items]

c. Mu aresi pu ime mov.
 me.DAT like.NPST that be.1SG.NPST purple
 ‘I like that I am purple.’

Condition 8: Would like/ X-marked predicate [6 items]

d. Tha mu arese na imun mov.
 MOD me.DAT like.PST.IPFV na.SUBJ be.1SG.PST.IPFV purple
 ‘I would like to be purple.’

30 typically developing monolingual Greek children, 4–9-years old (mean age: 7;7) participated in the study. To be able to observe the developmental path and age effect differences we divided participants into two age groups: Group A (n=15; age range: 4-7;11/ mean age: 6;4) and Group B (n=15; age range: 8-10/ mean age: 8;9). The split was decided based on the developmental shift we observed in children’s responses on the X-marked conditions. The control group consisted of 30 adults (mean age: 34). All participants were recruited privately.

6.3. Results of Experiment I

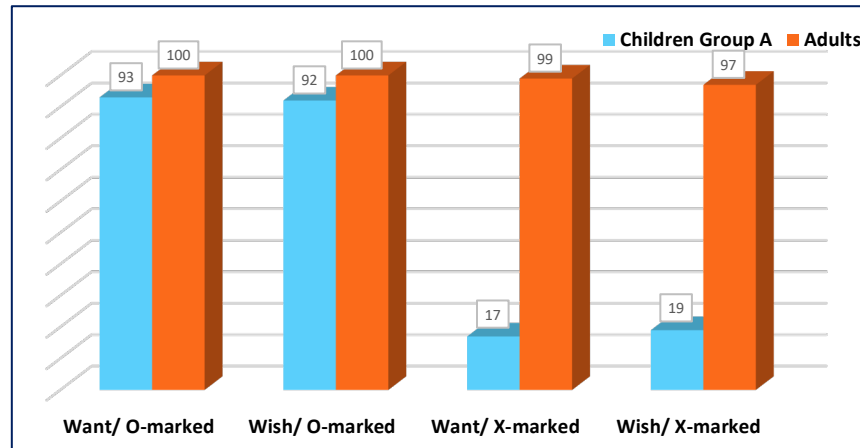


Figure 5: % of correct responses per condition for Group A

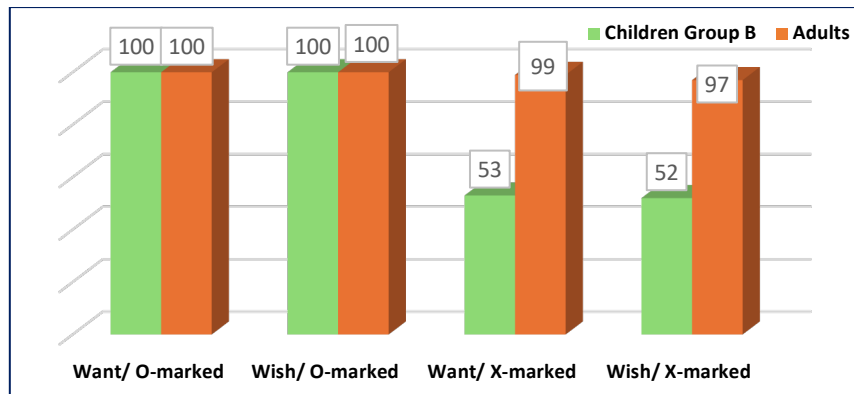


Figure 6: % of correct responses per condition for Group B

The results show that all children ($n=30$, up-to-date) perform adult-like on O-marked desires (Fig. 5 & Fig.6). Statistical analysis (2-way ANOVA) shows that younger children (Group A; $n=15$, mean: 6;4) fail to interpret X-marked desires up to a rate of 83% (for the verb *want*) and 81% (for the verb *wish*) (Fig. 5). Older children (Group B; $n=15$, mean: 8;9) perform significantly better ($p<0.001$) on X-marked desires, at a rate of 53% (Fig. 6). No significant difference was found regarding the type of the desire-verb, i.e. *thelo* ('want') versus *efhome* ('wish'), neither on the O-marked nor on the X-marked conditions within a group.

7. Experiment II: Comparing O-marked and X-marked desires to O-marked and X-marked conditionals

7.1. Procedure and Conditions

Experiment II aims at comparing O-marked desires and conditionals to X-marked desires and conditionals. The methodology is the same as in Experiment I (see Section 6.1). Given that we found no difference concerning the desire-verb tested in Experiment I, we decided to use the verb *thelo* (want) because it receives X-marking morphology when its complement is also X-marked.

As in Experiment I, children would watch pictures of the little creatures (as in Fig. 7 & Fig. 8) and they would listen to sentences as in (8).

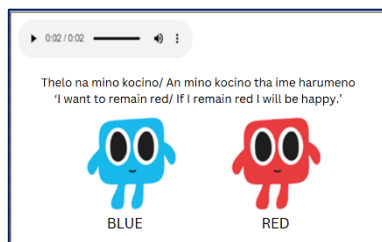


Figure 7: Example of O-marked desires and O-marked conditionals (Conditions 1 & 2)



Figure 8: Example of X-marked desires and X-marked conditionals (Condition 3 & 4)

(8) Condition 1: Want/ O-marked [6 items]

- a. Thelo na mino prasino.
 want.1SG.NPST.IPFV na.SUBJ remain.1SG.NPST.PERF green
 ‘I want to remain green.’

Condition 2: Conditional/ O-marked [6 items]

- b. An mino prasino, tha ime harumeno
 if remain.1SG.NPST.PERF green FUT be.1SG.NPST happy
 ‘If I remain green, I will be happy.’

Condition 3: Want/past X-marked [6 items]

- c. Tha ithela na iha mini prasino.
 MOD want.1SG.NPST.IPFV na.SUBJ remain.1SG.PLUP green
 ‘I wish I had remained green.’

Condition 4: Conditional/X- marked [6 items]

- d. An iha mini prasino, tha imun harumeno.
 if remain.1SG.PLUP green FUT be.1SG.PST.IPFV happy
 ‘If I had remained green, I would be happy.’

In total, after 3 warm up items for practicing, children had to answer to 24 randomized critical items (6 items/condition) and 24 fillers. The fillers are exactly the same as in Experiment I.

The participants of Experiment II were 14 typically developing monolingual Greek children, 6–9-year-old (n=14; age range:6-8/ mean age: 7;1). We targeted this age group for the reason that Group A in Experiment I performed poorly on X-marked conditions and we wanted to compare the participants in Experiment II to child Group B of Experiment I. The control group consisted of 20 Greek speaking adults (mean age: 34). All participants were recruited privately.

7.2. Results of Experiment II

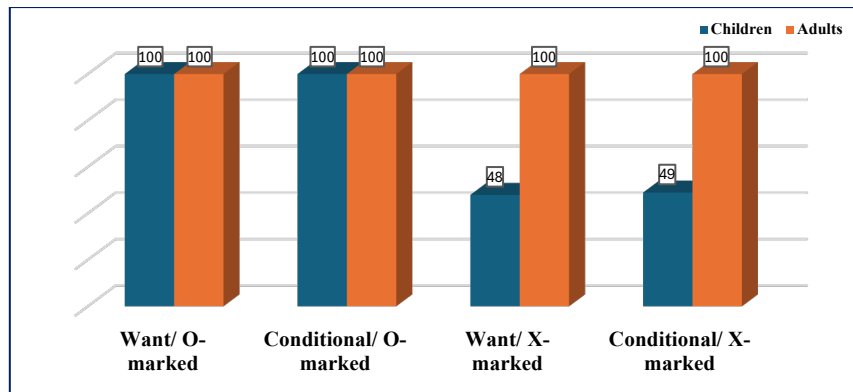


Figure 9: % of correct responses per condition

Figure 9 summarizes the results of Experiment II. All children perform adult-like (i.e. ceiling performance) on O-marked desires and O-marked conditionals and they fail to correctly interpret X-marked desires up to a rate of 52% and X-marked conditionals at a rate of 51%. No significant difference was found between X-marked desires and X-marked conditionals. In this respect, our findings suggest that X-marking encoding counterfactuality emerges at the same period irrespectively of the specific meaning (i.e. desires versus conditionals). Looking into the individual responses, we observe the same pattern in both constructions, namely that children at this age are split into those who perform adult-like in all conditions and the ones that make a significant number of misinterpretations in both X-marked desires and conditionals.

The experimental design raises an important question which calls for an answer. How can we be sure that children perform adult-like in O-marked conditions and it is not the case that their performance results from a go-for-the-color-you-hear strategy? The answer to this question comes from children's performance on the filler items.

Figure 10 represents children's overall performance in fillers. We observe that children's correct responses to O-marked conditions (i.e. Condition 5: *like/ O-marked predicate* and Condition 7: *like/ O marked activity*) range from 93-100%, similarly to O-marked conditions in Experiment I and II. The labels for the groups depicted in Fig.10 (i.e. Young-mixed and Old-mixed) refer to the sum of younger and older children respectively from both experiments since the fillers were the same across experiments. The age division between the two groups was at 8 years old. Adults' responses (not depicted in the graph) were 99%-100% correct across conditions, similarly to the critical items presented in Fig. 5, Fig. 6 and Fig. 9.

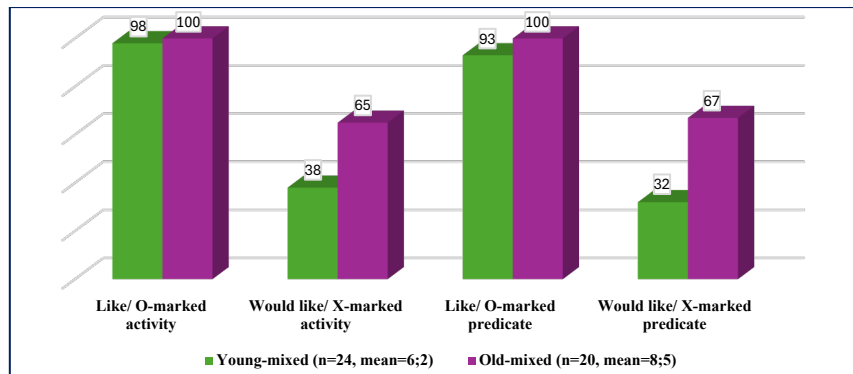


Figure 10: % of correct responses per condition in the fillers

Turning to the insight that fillers can provide regarding the X-marking conditions, we observe that children's correct responses to Condition 6 (*would like/ X-marked be-predicate*) and Condition 8 (*would like/ X-marked activity verb*) are comparable to X-marked conditions of the critical items in Experiment I and II. The group of young-mixed children (n=24, age range: 4-7;11/ mean age: 6;2) gave 38% and 32% correct responses to Conditions 6 and 8 respectively. The group of old-mixed children (n=20, mean = 8;5) performed significantly better than younger children, but still not in an adultlike way. The overall results in the filler-control items do not deviate from the bigger picture and shows that the responses to critical items were not driven by colour association.

9. Overall conclusions & Discussion

The overall findings suggest that younger children (up to the age of 7) fail at large to distinguish between O-marked and X marked desires; children tend to interpret X-marking as O-marking. Moreover, Experiment II has shown that children at similar ages fail to correctly interpret X-marked conditionals at an equal rate to X-marked desires. In this respect, our findings cannot provide a conclusive answer to research question (iii), namely, if the development of counterfactual conditionals precedes or follows the development of counterfactual desires. Our data seem to support the hypothesis that the two constructions emerge simultaneously, but we cannot exclude the possibility that we did not detect the stage at which the two constructions emerge distinctly; data from more participants may reveal significant differences in their development paths.

Approximately at the age of 7;6, a shift towards the adult-grammar is detected, despite the fact that some of the older children (Group B; mean age: 8;9) still produce a number of errors in the relevant conditions. This stage in particular, is characterised by a polarity performance: individual responses show that after the age of 7, children start displaying a split pattern and they can be divided into those who perform almost perfectly and those who still produce a large number of errors. This pattern was also detected in the Rouvoli, Tsakali & Kazanina

(2019) study on counterfactual conditionals in English; children's difficulty with counterfactual interpretation holds equally for X-marked desires and X-marked conditionals while the reverse pattern is also true: children who perform well on X-marked desires, do so on X-marked conditionals as well. This pattern suggests that the age at which children perform adult-like on counterfactuality may vary significantly.

It remains an open question at what age exactly pastX-marked structures have fully developed. Our findings definitely show that the development of X-marking is a prolonged developmental process, confirming previous research on counterfactual reasoning, which often assumed its completion by the age of 12 (Beck, Robinson, Carroll & Apperly 2006, Rafetseder, Schwitalla & Perner 2013).

However, whether children's preference for interpreting X-marked desires as O-marked is (partly) attributed to the Actuality bias (de Villiers 2005, Kazanina, Baker & Seddon 2020), and/or to immature counterfactual reasoning up to the age of 12 (Beck, Robinson, Carroll & Apperly 2006, Rafetseder, Schwitalla & Perner 2013), we believe that the development of the linguistic form (i.e. pluperfect) which is also used to encode counterfactuality in many languages is the key for the development of counterfactuality overall (see also Rouvoli, Tsakali & Kazanina (2019)). To our mind, the missing key in the up-to-date research in order to understand the development of counterfactuality goes through the acquisition of the pluperfect as temporal past, distinct from its morphological identification to counterfactuality.

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